International Journal of Environmental Impacts
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Sustainability Assessment of Water Treatment Plants Using Integrated Water Quality Index and Life Cycle Assessment

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Received: 12-11-2025 **Revised:** 04-02-2026 **Accepted:** 06-25-2026

Citation: A. K. Abdulwahd, R. Alsultani, and M. C. Liejy, "Sustainability assessment of water treatment plants using integrated water quality index and life cycle assessment," *Int. J. Environ. Impacts.*, vol. 9, no. 4, pp. 926–939, 2026. https://doi.org/10.56578/ijei090407.

Abstract: Treating drinking water is important in safeguarding human health through the elimination of contaminants and ensuring quality of water is high. Nevertheless, such a process is very likely to consume a lot of energy, chemicals, and complicated technologies that can increase the costs of production and lead to secondary effects on the environment. This paper is a combination of water quality index (WQI) and life cycle assessment (LCA) in assessing the environmental consequences of three water treatment plants in the Babil Governorate, Iraq: Al-Hilla, Al-Mahaweel, and Al-Hashimiyah. The WQI, which is determined by twelve fundamental water quality variables using the Canadian Council of Ministers of the Environment (CCME) approach, indicates that treated water is rated as good to all the three plants with the values of 94.719, 94.622, and 95.719. Using SimaPro 7.0 software and the Eco-Indicator 99 database, a LCA was performed that assessed the impact into categories such as global warming potential (GWP), acidification potential (AP), eutrophication potential (EP), human toxicity potential (HTP), energy consumption, chemical use and waste generation. It was found that all the water treatment plants reached Iraqi water quality standards with different environmental effects. The treatment plant that had the highest impacts in most of the categories was the Hashimiya treatment plant implying that it is balanced in both high water quality and high environment footprint. The given research identifies the necessity to combine WQI and LCA approaches to make the water treatment processes more sustainable and effective in order to contribute to the creation of the environmentally responsible water management policies in Babil Governorate and other areas.

Keywords: Water quality index; Life cycle assessment; Water treatment plant; Canadian Council of Ministers of the Environment approach; Simapro-Eco-Indicator 99; Environmental impact; Sustainability

1 Introduction

It is a basic need of the population to have access to safe and clean drinking water. Physical, chemical and biological contaminants present in water are dangerous to human health, and therefore, water treatment is necessary [1]. Nevertheless, it is costly in terms of energy, chemicals, and technologies, and this aspect raises the cost of the products produced and may affect the environment in other ways [2, 3]. Over the past few years, water treatment process evaluation has taken a new dimension whereby besides the conventional performance measures, environmental sustainability measurements are also considered. Such a turn indicates an increased sense of connection between human and environmental health and the sustainability of water resources over the long term [4].

The water quality index (WQI) and the life cycle assessment (LCA) are among the most frequent instruments to use in this overall assessment as they are used to measure the efficiency of the water treatment process and its environmental impact [5, 6]. WQI is a holistic measure of water quality incorporating many physical, chemical and biological aspects. These parameters usually have measures like pH, dissolved oxygen, turbidity, total dissolved solids (TDS), and contaminants like heavy metals and organic components [7]. The Canadian Council of Ministers of the Environment (CCME) approach is one such model among other WQI models that is highly esteemed due to its scientific rigor and reliability [8].

The WQI of the CCME uses a three-step approach, which includes identification of relevant water quality parameters, comparing the parameters against the set water quality standards, and the index is calculated with the help of a formula that takes into account the range, frequency, and magnitude of the exceedances [9, 10]. With this holistic model, the index of water quality obtained will be the true index of water quality in the eyes of the people, with both scientific and practicality [11–13].

It has been proven that the CCME WQI is an effective tool in summarizing different types of water quality data into one ratio that can be easily understood, which is necessary when managing water resources [14–16]. Its application is extensive across fields, including the evaluation of agricultural runoff and its effects on freshwater systems as well as the evaluation of the operation of urban water treatment plants [17]. Integrating the concept of life cycle analysis and the process of water treatment has made it possible to determine areas where the negative effects on the environment could be prevented, enhancing the overall sustainability of the water treatment processes [18].

The combination of life cycle analysis with the water treatment processes will help identify the opportunities to enhance the environmental performance. On the example of energy utilization during the treatment, environmentally friendly chemicals, and efficient water distribution system might help to decrease the total environmental impact of water treatment plants by a large margin [19]. Moreover, the LCA may be utilized to compare various treatment technologies and approaches and aid in choosing the most sustainable versions [20, 21]. LCA also helps to create effective and eco-friendly water treatment that improves the environment by pointing out the areas that need adjustments [22, 23].

Although WQI and LCA have been extensively used in the water management research, these two tools are usually used separately and therefore, they cannot be used to effectively assess both the performance of the treatment process and environmental sustainability. The originality of the present research is in the fact that the integrated WQI-LCA framework is developed, which evaluates the water quality enhancement and other environmental effects simultaneously. Contrary to the earlier studies, this one provides a direct correlation between the WQI results and the LCA indicators, which makes it possible to determine the trade-offs between the high quality of water and the reduction of the environmental burden. Moreover, the study offers one of the pioneer attempts to use this combined method of water treatment plants in Babylon Province, Iraq, on the basis of a broad set of data obtained during 16 months. The results provide useful information regarding the maximization of treatment procedures and sustainable management of water resources in data-vacuous areas.

2 Methodology

2.1 Description of the Study Area

The research location (Figure 1) is on three critical water treatment plants in Babil Governorate, Iraq: Al-Hilla, Al-Mahaweel, and Al-Hashimiyah as revealed in Figure 1a. These plants are important in supplying clean drinking water to a great number of people, guaranteeing the community good health and well-being [2].

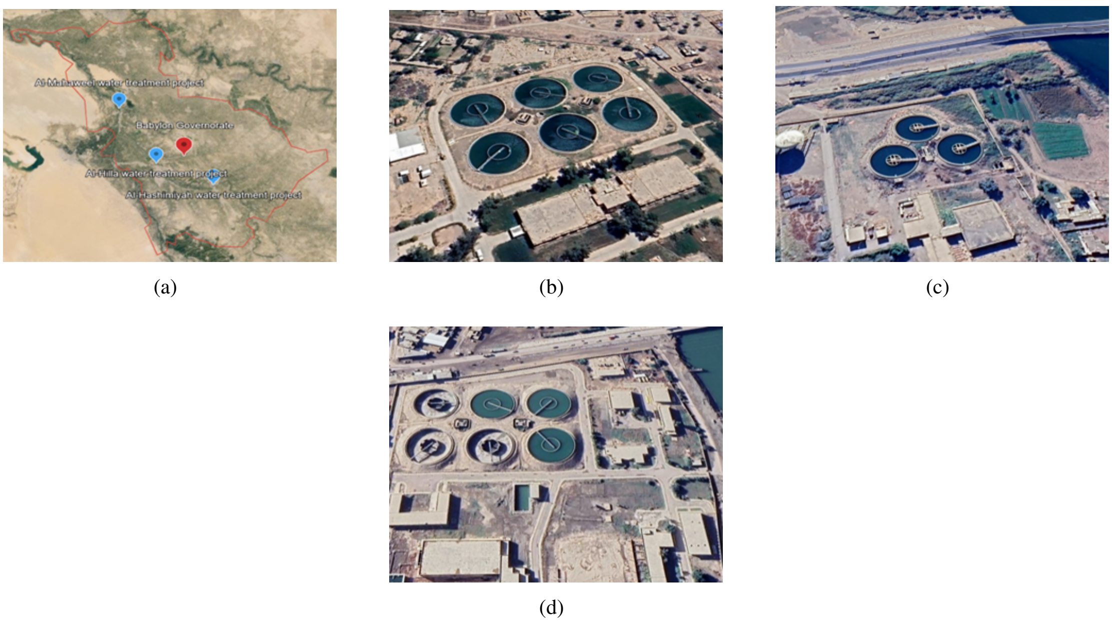

Figure 1. Google Earth map of selected water treatment plants: (a) study area overview; (b) Al-Hilla Water Treatment Plant; (c) Al-Mahaweel Water Treatment Plant; (d) Al-Hashimiyah Water Treatment Plant

Babylon Governorate has several challenges to water resources such as agricultural run-off, release of untreated or partially treated water, and industrial activities, which lead to the degradation of the quality of raw water. Besides, seasonal changes in the flow of rivers and weather conditions also influence the availability and quality of water, making the treatment process and the operational requirements more complex.

Al-Hilla Water Treatment Plant is located in the center of Babil Governorate (Figure 1b) and it is located in the city of Al-Hilla and its immediate environment. The plant has all the problems associated with cities, such as intensive use of drinking water, fluctuations in water supplies, and the risk of pollution by the application of industrial and agricultural enterprises [24]. The water treatment plant is situated in the southeastern section of the Babil Governorate (Figure 1c) and supplies clean water to the people of Al-Mahaweel and others located further. Among operational issues, there are seasonal changes in water quality and the need to streamline treatment procedures to easily eliminate pollutants and guarantee quality water supply.

The Al-Hashimiyah Water Treatment Plant is in the north of the Babil Governorate (Figure 1d) and it serves the city of Hashimiya and other people around it. The challenges that this plant has to deal with are peculiar, including the reduction of water quality due to agricultural runoff of nearby farms and the elimination of pollutants produced by local industries [22].

All water treatment plants are in different geographic, social, and economic settings, and it has a considerable influence on water sources utilized in treatment technologies and compliance with regulatory standards [25]. It is important to understand these factors in order to evaluate the efficiency of the processes of water treatment and the consequences of these processes on the health of people and the sustainability of the environment.

2.2 Samples Collection and Preservation

Raw and treated water samples were collected systematically in three sampled water treatment plants during 16 months, between March 2024 and June 2025. Sampling campaign was done monthly making the total number of sampling events of each plant 16. Water samples were sampled at both influent (raw water) and effluent (treated water) in every sampling event. Three replicate samples were taken on each type of water in each plant to get the reliability of data and reduce sampling errors. Based on this, the total sample collected was: 3 plants, 16 months, 2 types of water, 3 replicates = 288 samples. The sampling protocol was based on international standards to achieve accuracy and consistency [26, 27].

Sampling was performed at specific intervals to record seasonal changes and changes in water quality based on operation. As soon as the samples were collected, they were stored and treated in the usual way to avoid contamination and preserve the stability of parameters. The preservation was done by acidification (metal analysis) and refrigeration (4 °C) of physicochemical and microbiological parameters [28]. All samples were taken to a certified lab to be thoroughly analyzed in terms of physical, chemical, and biological values.

The analytical procedures were performed according to the globally accepted guidelines and national regulations [29, 30]. Strict quality assurance and quality control (QA/QC) measures were also performed to make sure that the results were reliable and accurate. These were use of certified reference materials, regular calibration of the instruments, blank samples and duplicated testing together with periodic proficiency tests [31]. Table 1 presents the selected water quality parameters used in the calculation of the WQI.

Table 1. Specifications of studied indicators under Iraqi treated water standards [2]

Indicator	Unit	Iraqi Standards for Water Quality
Biological oxygen demand (BOD)	mg/L	<40
Total dissolved solids (TDS)	mg/L	1000
pH	–	6.5–8.5
Electrical conductivity (EC)	μS/cm	1000
Turbidity	NTU	5
Potassium (K^+)	mg/L	10
Sodium (Na^+)	mg/L	200
Magnesium (Mg^{2+})	mg/L	30
Sulfate ($SO_4{}^{2-}$)	mg/L	250
Calcium (Ca^{2+})	mg/L	150
Total hardness (TH as $CaCO_3$)	mg/L	500
Temperature	°C	≤35

Note: A dash (–) denotes dimensionless.

2.3 Water Quality Index Calculations

The National Sanitation Foundation Water Quality Index (NSFWQI) which is also referred to as the Brown Index was created and is now being used by the U.S. national sanitation foundation [32], as a simple measure of water quality. The index has a range of 0 to 100 and it is a decreasing index since its numbers reduce as the water pollution increases. This index incorporates a holistic scale of 12 well-considered water quality parameters with a defined weight factor that acts as its significance in its contribution to human health. These indicators often encompass the level of pH, the dissolved oxygen level and turbidity, as well as the concentration of many pollutants, including heavy metals and organic ones [33–35].

NSFWQI is calculated as a product of the individual quality indices of each parameter multiplied by the coefficients of each:

$$\text{NSFWQI} = \prod_{i=1}^{n} \text{WQI}_i^{w_i} \tag{1}$$

where, NSFWQI is the overall National Sanitation Foundation Water Quality Index; WQI_i is the quality sub-index (rating) corresponding to the ith water quality parameter; w_i is the normalized weighting factor assigned to the ith parameter, reflecting its relative importance in the overall index; nis the total number of water quality parameters considered in the calculation; and idenotes the parameter index, ranging from 1 to n. The weighting factors satisfy the normalization condition $\sum_{i=1}^{n} w_i = 1$. This formulation follows the standard NSFWQI methodology, in which parameters with greater influence on overall water quality are assigned higher weighting factors.

Besides the NSFWQI, monthly testing of the raw and treated water quality was taken in relation to the Iraqi water quality standards using the CCME index methodology. One of the most famous models is the CCME WQI that consists of three major elements, namely, scope (F1), frequency (F2), and amplitude (F3) that are explained by Al-Janabi et al. [36].

The formula for computing standard CCME WQI is:

$$\text{CCME WQI} = 100 - \frac{\sqrt{\text{F}_1^2 + \text{F}_2^2 + \text{F}_3^2}}{1.732} \tag{2}$$

where,

$$\text{F1} = \left(\frac{\text{Number of failed variables}}{\text{Total number of variables}}\right) \times 100 \tag{3}$$

F2 indicates frequency with which the limitations are not met.

$$\text{F2} = \left(\frac{\text{Number of failed tests}}{\text{Total number of tests}}\right) \times 100 \tag{4}$$

F3 reflects the amount, as determined by the following formula, by which failed tested values are not in compliance with their goals (limits).

The deviation computed using Eq. (5) in cases when the test result cannot exceed the goal:

$$\text{Excursion } i = \left(\frac{\text{Failed test value } i}{\text{Objective } j}\right) - 1 \tag{5}$$

or from Eq. (6), where the test value is not less than the objective:

$$\text{Excursion } i = \left(\frac{\text{Objective } j}{\text{Failed test value } i}\right) - 1 \tag{6}$$

(i) By summing the individual test deviations from their objectives and dividing them by the total test number (all tests). The normalized sum of deviations (NSE), which reflects the collective quantity by which non-agreed individual tests are calculated was computed as follows:

$$\text{NSE} = \frac{\sum_{i=1}^{n} \text{Excursion } i}{\text{Total number of tests}} \tag{7}$$

where, NSE is the normalized sum of excursions; Excursion i is the excursion value for the ith failed test, representing the extent to which the measured value exceeds the prescribed water quality objective; n is the total number of failed tests for which excursions are calculated; and total number of tests is the total number of water quality measurements included in the assessment. The NSE value quantifies the collective amount by which failed tests deviate from their corresponding water quality objectives and is subsequently used in the calculation of the CCME WQI.

(ii) F3 can be calculated as:

$$\text{F3} = \frac{\text{NSE}}{\text{NSE} \times 0.01 + 0.01} \tag{8}$$

After the CCME WQI value was calculated, water quality was classified by linking it to the classes listed in Table 2.

Table 2. CCME WQI classification criteria [37, 38]

CCME WQI Value	Water Quality
95–100	Excellent
80–94	Good
65–79	Fair
45–64	Marginal
0–44	Poor

Note: CCME WQI, Canadian Council of Ministers of the Environment Water Quality Index.

This type of classification is an effective tool of evaluating the state of water quality and successfully implementing the strategies and interventions aimed at safeguarding the human health and environmental sustainability.

2.4 Life Cycle Assessment Approaches

LCA is a methodology that is systematic in the process of assessing the environmental effects of all the phases in the life cycle of a system, including the raw material extraction and ultimate disposal, in compliance with the ISO 14040 [39] and UNEP/SETAC [40] guidelines. One of the key elements of LCA is a clear definition of system boundary and functional unit [41, 42]. This paper has assumed a cradle-to-gate system boundary that would provide a thorough but narrow environmental analysis. The system boundary incorporates [43]:

• Upstream activities: The manufacturing and delivery of treatment chemicals (e.g., coagulants, disinfectants), and electricity generation.

• Basic operational activities: All on-site operations in the water treatment plants such as coagulation, flocculation, sedimentation, filtration, disinfection and sludge management.

• Outputs: Air, water, soil emissions and solid waste produced during treatment.

Nevertheless, the system boundary was not constituted by infrastructure construction, equipment manufacturing, and water distribution systems because of data constraints and comparatively small contribution of the environmental impact of operations on a study period. The functional unit was determined to be 1 m^3 of treated drinking water and this gave a consistent level of comparison among the three water treatment plants.

The LCA has been done under SimaPro 7.0 software with Eco-Indicator 99 method, using both the midpoint and endpoint impact categories, including global warming potential (GWP), acidification potential (AP), eutrophication potential (EP) and human toxicity potential (HTP). The LCA model has three key stages:

1. Life cycle inventory (LCI): List of all inputs (energy, chemicals) and outputs (emissions, waste) of the treatment processes.

2. Life cycle impact assessment (LCIA): Measurement of the environmental impacts in terms of standardized indicators.

3. Interpretation: Assessment of findings to determine significant environmental hotspots and areas where the sustainability can be enhanced. This is a well-defined system boundary that enhances consistency, transparency, and comparability of environmental impacts among the chosen water treatment plants.

2.5 Integrated Water Quality Index and Life Cycle Assessment Methodology

This research paper uses a comprehensive methodological approach in the assessment of water quality performance and environmental effects of the chosen water treatment facilities. The general work process is divided into four phases. The samples of water (raw and treated) were systematically collected within the three chosen plants in a period of 16 months and the major parameters of physicochemical and biological parameters were analyzed in the laboratory. Second, the WQI was determined by the CCME approach to determine the water quality conditions during pre and post treatment. Third, the SimaPro 7.0 with the Eco-Indicator 99 method was used to carry out a LCA to estimate the environmental impacts of the operational phase of the treatment plants, such as energy use, chemical consumption, and emissions. Lastly, statistical tests, such as descriptive statistics and Analysis of Variance (ANOVA), were conducted to compare the differences between the plants and find out the significant differences in the categories of environmental impact.

The LCA was carried out using the SimaPro software and background inventory data were mostly taken from the Ecoinvent database, which consists of data on the generation of electricity, manufacturing of chemicals and transportation processes. Operational data were collected from the water treatment plants studied and adapted to local conditions in Iraq for the use of these datasets. The Eco-Indicator 99 method was chosen because it had been used in many environmental assessment studies and was able to be evaluated in a consistent method.

This framework (Figure 2) allows the assessment of the treatment efficiency and environmental sustainability, identification of the possible trade-offs between the two, and the integration of WQI and LCA.

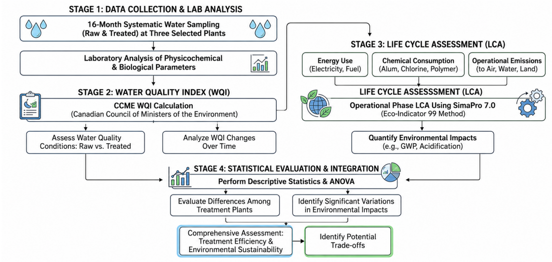

Figure 2. Integrated water quality index and life cycle assessment (WQI-LCA) methodology flowchart

3 Water Quality Index Results

It is evident that the estimated values of the chemical and physical characteristics of the analyzed water treatment facilities all fall within the Iraqi requirements except the calcium, turbidity, electrical conductivity (EC) of raw water, and temperature of processed water, as shown in Figure 3.

Figure 3. Raw water chemical and physical laboratory indicators for Al-Hilla, Al-Mahaweel and Al-Hashimiyah projects: (a) pH; (b) temperature; (c) turbidity; (d) electrical conductivity (EC); (e) total hardeness (TH); (f) calcium (Ca^{2+}); (g) magnesium (Mg^{2+}); (h) sulfate ($SO_4{}^{2-}$); (i) total dissolved solids (TDS); (j) sodium (Na^{+}); (k) potassium (K^{+}); (l) biological oxygen demand (BOD)

The WQI of the untreated and treated water was analyzed in compliance with the Iraqi water quality standards and regulations, through the CCME methodology.

F1 (scope) represents the percentage of monitored water quality parameters that failed to meet the Iraqi drinking water standards at least once during the monitoring period and was calculated using the same 12 parameters for both raw and treated water. F2 (frequency) expresses the percentage of individual water quality tests that did not comply with the prescribed standards relative to the total number of tests conducted at each sampling site. F3 (amplitude) quantifies the extent to which failed test results deviate from the corresponding regulatory objectives by calculating individual excursions, normalizing their cumulative effect as the NSE, and converting it into the amplitude factor. Together, these three factors were combined using the CCME WQI equation to evaluate the overall water quality status of each treatment plant.

Table 3 and Figure 4 show the summary results of CCME WQI and the general water quality classification. In the case of raw water, the values of WQI were 81.232 (Al-Hilla), 79.308 (Al-Mahaweel), and 80.932 (Al-Hashimiyah), and these values refer to a fair water quality, respectively. It is explained by high concentration of such parameters as turbidity, calcium and EC that exceeds the admissible levels, and anthropogenic factors, like agricultural runoff and sewage discharge, as earlier indicated by Ethaib et al. [25] and Sener et al. [44]. The quality of the treated water also improved significantly, with Al-Hilla, Al-Mahaweel and Al-Hashimiyah having a WQI value of 94.719, 94.622, and 95.719, respectively, which is indicative of good to excellent water quality conditions. These gains indicate successful treatment performance in spite of environmental pressures like soil percolation, rainfall intensity and variability in river flow [25, 44]. Table 4 also outlines the comparative outcomes and temporal difference of the quality of raw and treated water. The research study, which was carried out in the period between March 2024 and June 2025, involved descriptive statistical analysis (mean values) to determine the differences in the parameters of water quality in the three stations (Al-Hilla, Al-Mahaweel, and Al-Hashimiyah). In terms of specific parameters, the biological oxygen demand (BOD) levels in the raw water reveal that there is moderate level of organic pollution in all sites, but the total absence of BOD in the treated water is the evidence of the high level of the treatment processes. TDS and EC exhibited some spatial changes in raw water but small changes in the end of treatment, which suggest the maintenance of mineral balance. Likewise, pH, temperature, and turbidity showed small differences in the samples of raw water, which indicates the relatively similar environmental conditions in all sites. The values after treatment were maintained at optimum levels, which indicated the maintenance of the physicochemical equilibrium. The major ions, such as K^+, Na^+, Mg^{2+}, Ca^{2+}, and $SO_4{}^{2-}$, were not significantly different in raw and treated water, which further proves the effectiveness of the treatment processes in keeping the ionic stability and also making the water meet the potable water standards.

Table 3. The stations' water quality categorization and F1, F2, F3, and CCME WQI readings

Stations	Parameter	Al-Hilla	Al-Mahaweel	Al-Hashimiyah
Raw water	WQI value	81.232	79.308	80.932
	Classification	Fair	Fair	Fair
	F1	25	25	25
	F2	17.592	21.296	17.592
	F3	11.050	14.351	12.501
Treated water	WQI value	94.719	94.622	95.719
	Classification	Good	Good	Excellent
	F1	9.090	9.090	8.200
	F2	1.010	2.020	1.010
	F3	0.158	0.296	1.550

Note: CCME WQI, Canadian Council of Ministers of the Environment Water Quality Index; F1, scope; F2, frequency; F3, amplitude.

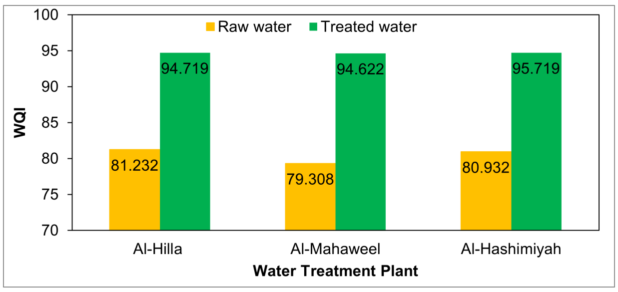

Figure 4. Water quality index (WQI) comparison selected treatment plants

Overall, the findings demonstrate that all three water treatment plants significantly improved water quality through effective contaminant removal, particularly of organic pollutants. While a few individual parameters did not consistently meet the recommended standards at all sites, the overall treated water quality showed substantial improvement compared with the raw water.

Table 4. Mean values of water quality indicators of selected water treatment plants

Characteristic	Al-Hilla		Al-Mahaweel		Al-Hashimiyah	
	Raw	Treated	Raw	Treated	Raw	Treated
Water quality index (WQI)	81.232	94.719	79.308	94.622	80.932	95.719
pH	7.3	7.5	7.6	7.5	7.3	7.3
Temperature (°C)	22.6	23.1	25.7	25.8	23.5	23.4
Turbidity (NTU)	11.2	0.8	13.9	2.9	12.4	2.6
Electrical conductivity (EC) (μS/cm)	1030.5	1032.6	999.3	1115.5	1053.9	1058.9
Total hardness (TH) (mg/L)	370	367	379.3	382.9	372.1	372.8
Ca^{2+} (mg/L)	82.8	82.1	84.7	89	85.2	86.5
Mg^{2+} (mg/L)	37.4	37.2	37.3	38.1	39	39.7
SO_4 (mg/L)	253.1	249.7	256	260.3	253.4	251.7
Total dissolved solids (TDS) (mg/L)	623.9	623.8	647.5	653.6	638.3	658.7
Na^+ (mg/L)	76.5	76.5	82.9	86.3	78.2	77.4
K^+ (mg/L)	3.7	3.7	4.1	4.1	3.8	3.7
Biological oxygen demand (BOD) (mg/L)	0.4	0	0.3	0	0.3	0

4 Life Cycle Assessment Results

LCA of Al-Hilla, Al-Mahaweel and Al-Hashimiyah Water Treatment Plants gives insights on the impacts that the three sites have on the environment based on the various indicators. SimaPro 7.0 software and Eco-Indicator 99 database were used to assess the impact based on various categories.

4.1 Environmental Impact Categories

The LCA results, presented in Figure 5, reveal notable variations in environmental impacts among the three water treatment plants. Al-Hilla Water Treatment Plant demonstrated moderate environmental performance, with a GWP of 1.25 kg CO_2-eq/m^3, reflecting balanced energy consumption and emissions. In contrast, Al-Mahaweel Water Treatment Plant showed a lower GWP (0.35 kg CO_2-eq/m^3), while Al-Hashimiyah Water Treatment Plant recorded the highest GWP (1.45 kg CO_2-eq/m^3), indicating a significantly larger carbon footprint primarily driven by intensive energy use and operational emissions.

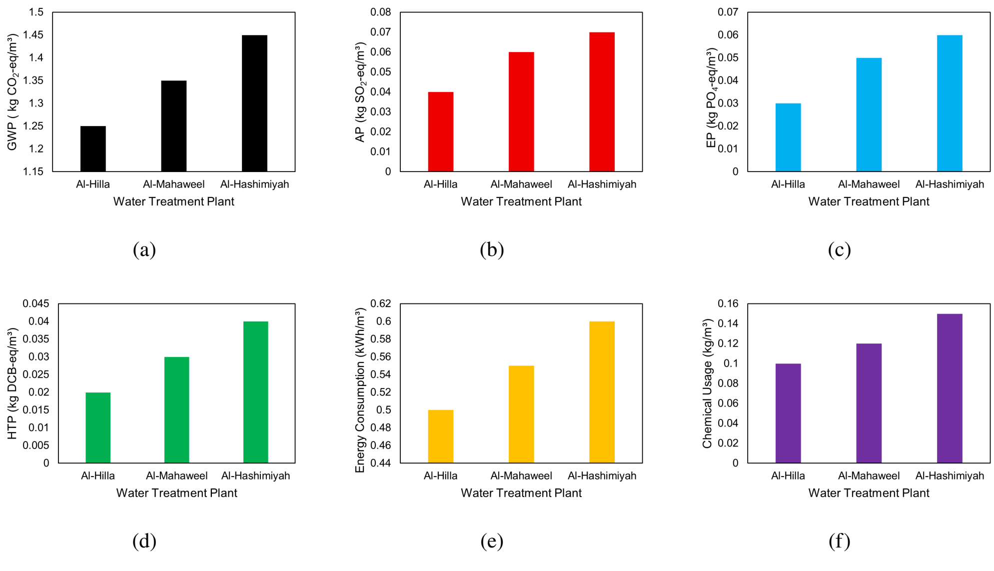

Figure 5. Life cycle assessment (LCA) results of environmental impact categories for the selected water treatment plants: (a) global warming potential (GWP); (b) acidification potential (AP); (c) eutrophication potential (EP); (d) human toxicity potential (HTP); (e) energy consumption; (f) chemical usage

A similar trend was observed for AP, where Al-Hilla Water Treatment Plant exhibited the lowest value (0.04 kg SO_2-eq/m^3), followed by Al-Mahaweel (0.06 kg SO_2-eq/m^3), and Al-Hashimiyah (0.07 kg SO_2-eq/m^3), suggesting increasing contributions to acidifying emissions. Regarding EP, Al-Hilla again showed the lowest impact (0.03 kg PO_4-eq/m^3), whereas Al-Mahaweel (0.05 kg PO_4-eq/m^3) and Al-Hashimiyah (0.06 kg PO_4-eq/m^3) indicated higher risks of nutrient enrichment and potential ecological imbalance.

HTP followed the same pattern, with values of 0.02, 0.03, and 0.04 kg DCB-eq/m^3 for Al-Hilla, Al-Mahaweel, and Al-Hashimiyah Water Treatment Plants, respectively, highlighting increasing concerns related to chemical exposure and toxic emissions.

From an operational perspective, Al-Hilla Water Treatment Plant exhibited the most efficient performance, with the lowest energy consumption (0.50 kWh/m^3), and chemical usage (0.10 kg/m^3). Conversely, Al-Hashimiyah Water Treatment Plant recorded the highest values in all operational indicators, energy consumption (0.60 kWh/m^3), and chemical use (0.15 kg/m^3), indicating more resource-intensive processes. Al-Mahaweel Water Treatment Plant showed intermediate performance across all indicators.

The overall integrated analysis of both WQI and LCA reveals the water treatment performance and environmental burden of the plants studied is very apparent. The majority of treatment plants experienced significant increases in the quality of the treated water, with the higher WQI's generally associated with increased water-use and environmental effects during plant operation. Out of water treatment plants, Al-Hashimiyah Water Treatment Plant has the highest water quality indicator of 94.719 which falls under the good category, corresponding to the best water treatment plant performance level in terms of its water quality indicator parameters which include energy consumption, chemical usage and environmental impact parameters (GWP, AP, EP and HTP). Similarly, Al-Mahaweel Water Treatment Plant had medium environmental and operational impacts and treated water quality performance. But, Al-Hilla Water Treatment Plant had relatively low environmental burdens, and yet had high treated water quality standards. These results suggest that better treated-water quality will be more expensive in terms of energy requirements and chemicals used, and thus more damaging to the environment. The results therefore highlight the importance of integration of the water quality targets with a viable operating practice to limit environmental impacts and water safety.

4.2 Statistics Analysis of Life Cycle Assessment Impacts

To give a sound statistical analysis of the LCA findings of the three water treatment plants, descriptive and inferential statistics were used.

The statistical analysis sample consisted of the operational records within 16 months (March 2024 to June 2025). Monthly values were taken into account in each category of environmental impact, and this made 16 observations per plant (n = 16) and 48 observations per all the plants. The resulting observations were obtained with respect to the measured LCA outputs (e.g., energy consumption, chemical use, and emissions) of the respective sampling period.

IBM SPSS Statistics (31.0.2.0) was used to conduct statistical analysis. The premises of the normality (Shapiro–Wilk test) and homogeneity of variance (Levene test) were checked and met before the inferential tests were conducted.

First, the descriptive statistics have been obtained to summarize the key features of the environmental impact data. The mean values are the average burden on the environment and the standard deviation is the variation of the data and range is the difference between the minimum and the maximum.

This was followed by a one-way ANOVA to identify the existence of statistically significant differences amongst the three water treatment plants of each of the environmental impacts categories. The analysis was performed to the confidence of 95% (α = 0.05).

The hypotheses that were tested were:

• Null hypothesis (H0): The three water treatment plants have no significant difference in terms of environmental impacts.

• Alternative hypothesis (H1) There exists a statistically significant difference in environmental impact in at least one water treatment plant.

In the categories of impacts where statistically significant differences were found ($p < 0.05$), post-hoc analysis with the Tukey Honestly Significant Difference (HSD) test was conducted to ascertain which specific pairs of water treatment plants differed significantly. A summary of the statistical results is presented in Table 5.

The results of the ANOVA as shown in Figure 6 show that EP and Energy Consumption had significant difference among the plants ($p < 0.05$). Conversely, GWP and AP were borderline significant indicating a moderate variation. The rest of the categories such as HTP, Chemical Usage and Waste Generation showed no statistically significant differences ($p > 0.05$) though variations are evident.

Comprehensively, the statistical analysis proves that all three water treatment plants have similar levels of performance in water quality, yet, their effects on the environment vary within certain categories, which leads to the opportunities of specific optimization and better sustainability.

Table 5. Statistical analysis results for environmental impact indicators among the three water treatment plants

Impact Category	Shapiro–Wilk (p-Value)	Levene (p-Value)	ANOVA F-Value	ANOVA p-Value
GWP	0.218	0.462	5.12	0.05
AP	0.341	0.527	4.87	0.07
EP	0.184	0.376	6.23	0.03
HTP	0.273	0.641	3.45	0.12
Energy consumption	0.156	0.288	7.01	0.02
Chemical usage	0.447	0.713	2.90	0.15
Waste generation	0.298	0.534	4.50	0.08

Note: ANOVA, Analysis of Variance; GWP, global warming potential; AP, acidification potential; EP, eutrophication potential; HTP, human toxicity potential.

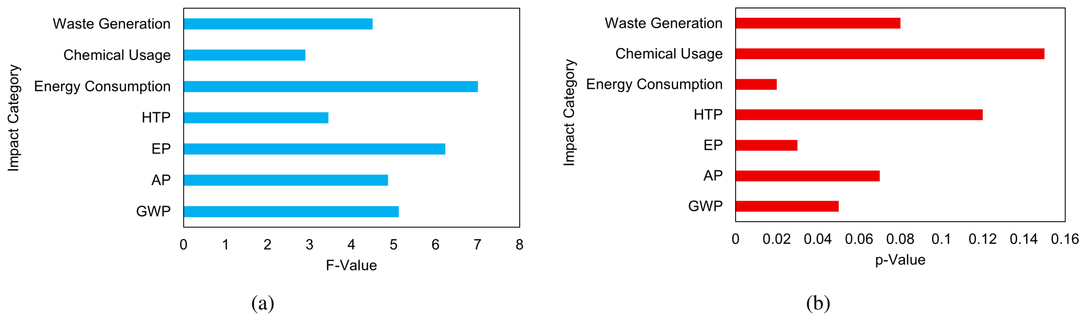

Figure 6. Analysis of Variance (ANOVA) results for environmental impact categories across water treatment plants: (a) F-value; (b) p-value

Note: GWP, global warming potential; AP, acidification potential; EP, eutrophication potential; HTP, human toxicity potential.

5 Environmental Impact Assessment

In this section, a comparative evaluation of the environmental performance of the Al-Hilla, Al-Mahaweel, and Al-Hashimiyah water treatment plants will be provided with a view to finding the possibilities of improving sustainability and operational effectiveness. According to the analysis that was performed with the help of SimaPro and Eco-Indicator 99, resources depletion, human health, and ecosystem quality are the most important aspects of the environment. The main contributors of these impact were found to be the energy consumption and emissions, with the level of energy consumption being the highest in Al-Hilla plant. Also, the use of chemicals especially during coagulation and disinfection processes was also discovered to pose a great impact on human health and quality of environment.

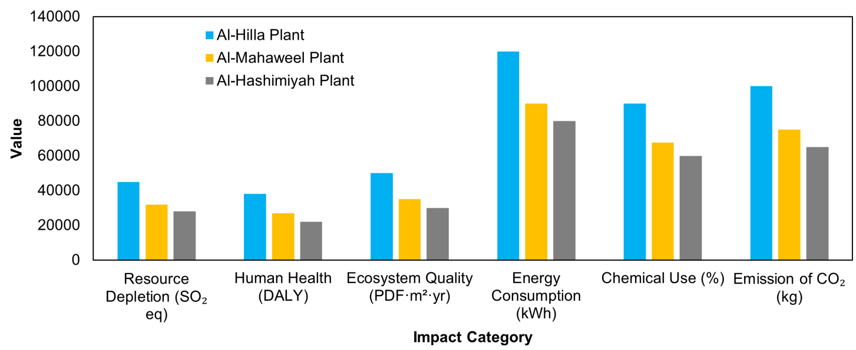

Figure 7. Results from environmental impact assessment using eco-indicator 99 of water treatment plants

Note: DALY, Disability-Adjusted Life Years; PDF, Potentially Disappeared Fraction.

Figure 7 has indicated that the opportunities to enhance sustainability in all plants are large. In the case of Al-Hilla, the priority steps can be the energy efficiency and the incorporation of renewable energy sources, like solar and wind power. Moreover, the negative health and ecological impact may be minimized by the use of chemicals and the adoption of environmentally friendly materials as it was adopted by Buzaubakova and Bedelbayeva [45]. Though the environmental impacts of Al-Mahaweel and Al-Hashimiyah plants are relatively lower, the energy efficiency and chemical management should be improved. In order to meet these objectives, the research suggests: (1) the use of energy-efficient technologies and the examination of renewable energy sources, (2) optimization of chemicals consumption and substitution with less toxic ones, and (3) the constant tracking of the environmental performance through the use of such programs as SimaPro. All of these strategic measures can contribute to a higher level of sustainability, increased efficiency of the work of the water treatment plants, and decreased environmental costs in general.

The resulting LCA results are roughly comparable to those for conventional drinking water treatment plants reported in other studies where the impacts mainly stem from energy use and chemical use. The values of the observed GWP, AP, EP, and HTP indicators are in the moderate range when compared to similar indicators from literature; however, differences exist between the studies as the treatment technologies, operational conditions, energy sources, system boundaries and regional characteristics vary. In particular, the higher the treatment efficiency, the higher the energy requirements and the larger the amount of chemicals used, the greater the environmental burdens that can be seen in the plants. Hence, the results of the present study are deemed to be sound and are comparable with other published water treatment LCA studies.

The indicators of environmental impact were measured based on a series of metrics in order to have a holistic analysis of the water treatment processes. The depletion of resources is measured using kilograms of sulfur dioxide (SO_2) equivalent, which is an indicator of how many resources are extracted and how it affects the environment. Disability-Adjusted Life Years (DALY) are used to quantify the human health impacts of the treatment activities, which may be associated with the potential disease burden. The Potentially Disappeared Fraction (PDF) of species per square meter per year is used to measure the quality of the ecosystem, which is a metric of the possible loss of biodiversity. In the measurement of energy consumption, kilowatt-hours (kWh) are used to denote the total energy demand of the plants. The qualitative nature of chemical usage is categorized as high, medium, and low, which represents the level of chemical usage and its possible environmental and health consequences. Lastly, the emission of carbon dioxide (CO_2) is measured in kilograms and it gives an approximation of the greenhouse gases that the plant activities produce.

6 Conclusions

This paper provides a depth analysis of water quality and environment conditions in the Al-Hilla, Al-Mahaweel, and Al-Hashimiyah Water Treatment Plants basing on the data of the period between mid-March 2024 and mid-June 2025. Key findings include:

1. Raw water in Al-Hilla (WQI = 81.232), Al-Mahaweel (WQI = 79.308) and at Al-Hashimiyah (WQI = 80.932) was categorized as fair WQI. High turbidity, calcium and EC at Al-Mahaweel, which is beyond regulations. These were attributed to the environmental and anthropogenic sources like the agricultural runoff and sewage.

2. The post treatment WQI values had considerable good improvement in all the plants: Al-Hilla (WQI = 94.719), Al-Mahaweel (WQI = 94.622) and Al-Hashimiyah (WQI = 95.719) which showed that the contaminants were removed successfully and the quality of water was improved.

3. The levels of BOD in raw water were stable, and the treatment of organic contaminants was successful, and the level of BOD in the treated water became zero.

4. TDS and EC values recorded slight differences between raw and treated water indicating that the treatment process is effective in ensuring that the mineral content is acceptable.

5. The pH values did not change after the treatment and the turbidity was reduced considerably, which may be regarded as the successful elimination of suspended particles.

6. Pre- and post-treatment of major ion levels (Ca^{2+}, Mg^{2+}, ${SO_4}^{2-}$, Na^{+}, and K^{+}) showed consistency, and this indicates the capacity of the treatment process to maintain the necessary ionic balance.

7. It was found that there were significant effects of GWP and AP. Although all the categories were not statistically significant evaluation, the optimization of the treatment measures might contribute to better environmental performance and reducing adverse effects.

8. The Al-Hilla, Al-Mahaweel, and Al-Hashimiyah Water Treatment Plants could be improved with specific energy efficiency measures along with the optimization of chemicals use, which would make them have a lower environmental impact and increase their sustainability and efficiency.

Finally, the paper confirms that the treatment regimes of the Al-Hilla, Al-Mahaweel, and Al-Hashimiyah Water Treatment Plants are very effective and as such, the quality of water has improved significantly. These changes in the scores of WQI and the decrease in the number of contaminants indicate the positive outcome of these processes. Nonetheless, to preserve high water quality standards, it is necessary to do continuous monitoring and optimization of the processes. The next line of research ought to be the optimization of the water quality indices and the investigation of the way to reduce the environmental harm and maintain the good water treatment mechanisms.

Author Contributions

Conceptualization, A.K.A. and R.A.; methodology, A.K.A. and R.A.; software, A.K.A.; validation, A.K.A., R.A., and M.C.L.; formal analysis, R.A.; investigation, A.K.A.; resources, A.K.A. and M.C.L.; data curation, A.K.A.; writing, original draft preparation, A.K.A.; writing, review and editing, R.A. and M.C.L.; visualization, A.K.A. and R.A.; supervision, R.A.; project administration, R.A.; funding acquisition, M.C.L. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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